

EXPERIMENT 7

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Student Name: Hunar Katyal

UID: 24BDA70085

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

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Subject Name: ADBMS

Subject Code: 24CSH-337

Experiment 6 – Understand Database Normalisation

Aim: To understand NF1, NF2, NF3, BCNF and problem solving using Relations and Function dependency.

Find Highest Normalisation form of the given Relations.

Problem 1: R(ABCDE)

$CE \rightarrow D$

$D \rightarrow B$

$C \rightarrow A$

Sol: Candidate key is CE

Relation not in BCNF as non-super key is determining non-prime key. $D \rightarrow B$

Relation not in 3NF as non-prime attribute determining non-prime attribute

Relation not in 2NF as subset of Candidate key is determining a non-prime attribute.

Hence Relation is in 1NF

Problem 2: R(ABCDEF)

$AB \rightarrow C$

$DC \rightarrow AE$

$E \rightarrow F$

Sol: Candidate key is ABD

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Relation is in 1NF as subset of candidate is determining non-prime attribute

Problem 3: R(ABCDEFGHI)

$AB \rightarrow C$

$BD \rightarrow EF$

$AD \rightarrow GH$

$A \rightarrow I$

Sol: Candidate key is ABD

Relation is in 1NF.

Subset of candidate key is determining non-prime attribute

Problem 4: R(ABCDE)

$AB \rightarrow CD$

$D \rightarrow A$

$BC \rightarrow DE$

Sol: Candidate Key is BC

Relation is in 2NF as partial dependency is removed and non-super key is determining the attribute

Problem 5: R(ABCDE)

$BC \rightarrow ADE$

$D \rightarrow B$

Sol: Candidate Key is BC, CD

Relation is in 3NF, non-super key is determining the attribute.

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Problem 6: R(ABC)

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow A$

Sol: Candidate key is A,B,C

Relation is in 3NF, prime attributes determining prime attribute but non-super key determining the attribute

Problem 7: R(F1, F2, F3, F4, F5)

$F1 \rightarrow F3$

$F2 \rightarrow F4$

$(F1, F2) \rightarrow F5$

Sol: Candidate key is (F1 F2)

Relation is in 1NF, subset of candidate key determining attribute.

Problem 8: R(ABCDEFGHI)

$AB \rightarrow C$

$AD \rightarrow GH$

$BD \rightarrow EF$

$A \rightarrow I$

$H \rightarrow J$

Sol: Candidate key is ABD

Relation is in 1NF as subset of candidate key determining attribute.

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Problem 9: R(ABCDE)

A $\rightarrow$ B

B $\rightarrow$ E

C $\rightarrow$ D

Sol: Candidate key is AC

Relation is in 1NF, subset of candidate key is determining the attribute.

Learning Outcome

After completing this experiment, the learner will be able to:

1. Understand Functional Dependencies.
2. Understand Database Normalisation forms.
3. Understand NF1, NF2, NF3, BCNF.